

Geometry & Measurement



Angle facts

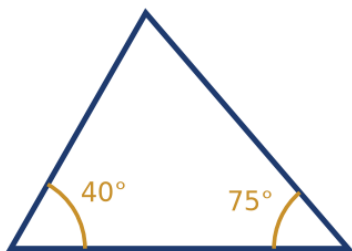
- 1 Two angles on a straight line measure $3x$ and $2x + 30$. Find x .
- 2 Find the angle vertically opposite a 65° angle.
- 3 Three angles around a point measure 110° , 95° , and x . Find x .
- 4 Two angles are complementary (sum to 90°). One measures 37° . Find the other.

Parallel lines and transversals

- 5 Two parallel lines are cut by a transversal, creating an angle of 70° . Find its corresponding angle.
- 6 In the same setup, find the co-interior (allied) angle to a 70° angle.
- 7 Find the alternate interior angle to a 55° angle.
- 8 Two parallel lines cut by a transversal create an angle of $4x$ and its co-interior angle of $2x + 30$. Find x .

Triangle angle sum

- 9 Find the third angle of the triangle shown, given the two marked angles.



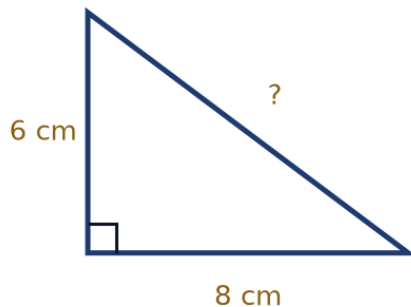
- 10 An isosceles triangle has two equal angles of 72° each. Find the third angle.
- 11 In a right triangle, one of the non-right angles is 35° . Find the other non-right angle.
- 12 A triangle has three equal angles. Find each angle, and name this type of triangle.

Angle sums of polygons

- 13 Find the sum of the interior angles of a hexagon (6 sides).
- 14 Find the sum of the interior angles of a pentagon (5 sides).
- 15 A regular polygon has an interior angle sum of $1,080^\circ$. How many sides does it have?

The Pythagorean theorem

- 16 Find the hypotenuse of the right triangle shown.



- 17 A right triangle has hypotenuse 13 and one leg 5. Find the other leg.
- 18 A 10 m ladder leans against a wall with its foot 6 m from the wall's base. How high up the wall does it reach?
- 19 A right triangle has legs 9 and 12. Find the hypotenuse.

Perimeter and area of rectangles and squares

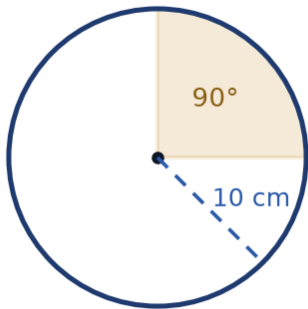
- 20 A rectangle has length 12 cm and width 7 cm. Find its perimeter and area.
- 21 A square has side length 9 cm. Find its perimeter and area.
- 22 A rectangle has perimeter 50 cm and length 16 cm. Find its width and area.
- 23 A square has area 121 cm^2 . Find its side length and perimeter.

Area of triangles and parallelograms

- 24 A triangle has base 14 cm and height 9 cm. Find its area.
- 25 A parallelogram has base 12 cm and height 8 cm. Find its area.
- 26 A triangle has area 60 cm^2 and base 15 cm. Find its height.
- 27 A parallelogram has area 126 cm^2 and height 9 cm. Find its base.

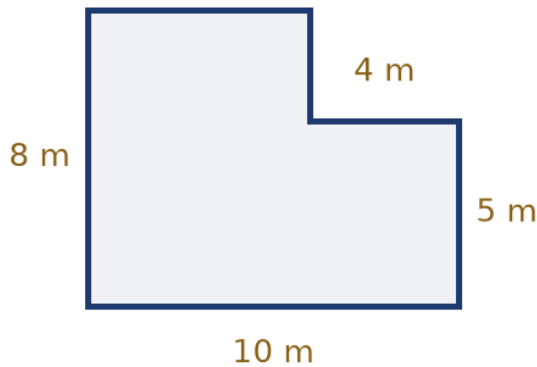
Circles: circumference and area

- 28 A circle has radius 7 cm. Find its circumference, using $\pi = \frac{22}{7}$.
- 29 A circle has radius 7 cm. Find its area, using $\pi = \frac{22}{7}$.
- 30 A circle has diameter 20 cm. Find its circumference in terms of π .
- 31 Find the area of the shaded 90° sector shown, in terms of π .



Composite figures

- 32 Find the area of the L-shaped room shown.



- 33 A garden is L-shaped: a 15 m by 6 m rectangle with a 5 m by 2 m rectangle cut from one corner. Find its area.
- 34 A shape consists of a 12 cm by 5 cm rectangle with a right triangle of base 6 cm and height 5 cm attached to one of the short sides. Find the total area.

Volume and surface area

- 35 A rectangular box has dimensions 5 cm $\times$ 4 cm $\times$ 3 cm. Find its volume.
- 36 A cube has edge length 6 cm. Find its volume and surface area.
- 37 A rectangular prism has volume 240 cm³, length 10 cm, and width 6 cm. Find its height.
- 38 Find the surface area of a rectangular box with dimensions 8 cm $\times$ 5 cm $\times$ 4 cm.

Word problems

- 39 A rectangular garden measuring 20 m by 15 m is surrounded by a path 1 m wide. Find the area of the path alone.
- 40 New flooring costs 45 riyals per square meter. Find the total cost to floor a rectangular room measuring 6 m by 4 m.
- 41 A 17 m ladder leans against a vertical wall with its base 8 m from the wall. How high up the wall does it reach?

- 42 A jogger runs 4 laps of a circular track of radius 35 m, using $\pi = \frac{22}{7}$. Find the total distance covered.
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MCQ — Angle facts

- 43 Two angles are supplementary. One is three times the other. Find the smaller angle.
- a. 60°
 - b. 30°
 - c. 135°
 - d. 45°
- 44 Three angles around a point measure 118° , 142° , and x . Find x .
- a. 80°
 - b. 140°
 - c. 120°
 - d. 100°
- 45 An angle is four times its complement. Find the angle.
- a. 54°
 - b. 18°
 - c. 72°
 - d. 90°
- 46 Two angles on a straight line measure $(4x + 10)^\circ$ and $(2x + 20)^\circ$. Find x .
- a. 20
 - b. 35
 - c. 15
 - d. 25

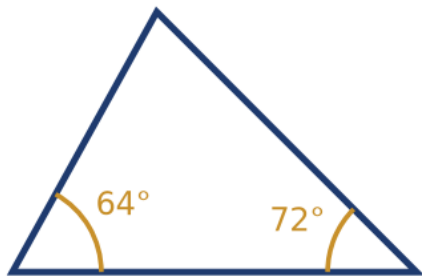
MCQ — Parallel lines and transversals

- 47 Two parallel lines are cut by a transversal. One angle is $(3x + 15)^\circ$ and its corresponding angle is $(5x - 25)^\circ$. Find x .
- a. 10
 - b. 8
 - c. 30
 - d. 20
- 48 Two parallel lines cut by a transversal create co-interior angles of $(2x + 10)^\circ$ and $(3x - 30)^\circ$. Find x .
- a. 30
 - b. 40
 - c. 50
 - d. 34
- 49 Two parallel lines are cut by a transversal. One angle is 63° . Find its alternate exterior angle.
- a. 153°
 - b. 27°
 - c. 63°
 - d. 117°

- 50 Two parallel lines are cut by a transversal. One angle and its co-interior angle differ by 40° and sum to 180° . Find the larger angle.
- a. 110°
 - b. 70°
 - c. 140°
 - d. 90°
-

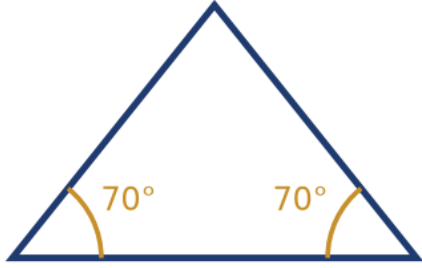
MCQ — Triangle angle sum

- 51 Find the third angle of the triangle shown.



- a. 116°
- b. 64°
- c. 54°
- d. 44°

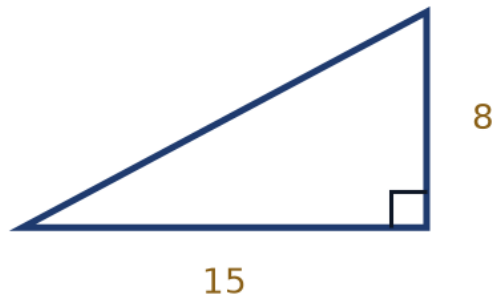
- 52 The isosceles triangle shown has two equal base angles of 70° each. Find the apex angle.



- a. 55°
 - b. 70°
 - c. 40°
 - d. 110°
- 53 A triangle has three angles in the ratio 2 : 3 : 4. Find the largest angle.
- a. 90°
 - b. 100°
 - c. 80°
 - d. 60°
- 54 In a right triangle, one non-right angle exceeds the other by 26° . Find the larger non-right angle.
- a. 32°
 - b. 58°
 - c. 64°
 - d. 48°

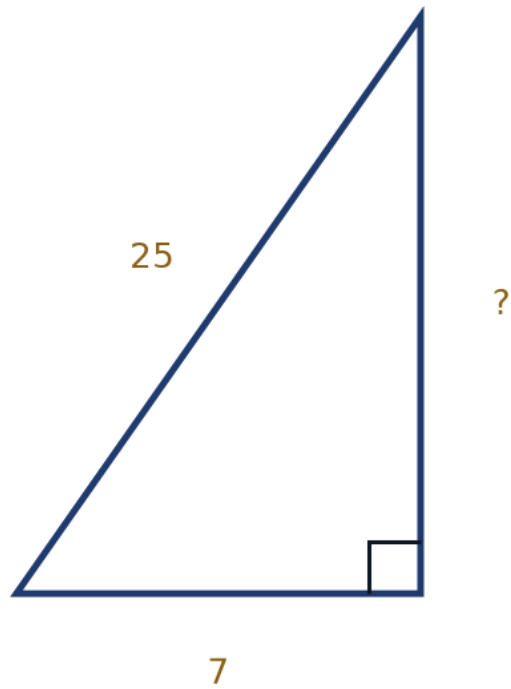
MCQ — The Pythagorean theorem

55 Find the hypotenuse of the right triangle shown.



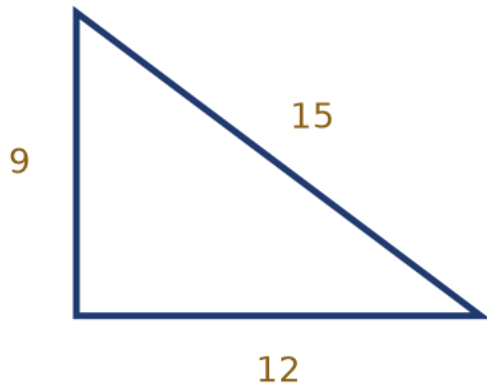
- a. 23
- b. 19
- c. 17
- d. 13

- 56 Find the missing leg of the right triangle shown.



- a. 26
 - b. 18
 - c. 24
 - d. 20
- 57 A rectangular field is 24 m long and 18 m wide. Find the length of its diagonal.
- a. 30
 - b. 36
 - c. 42
 - d. 26

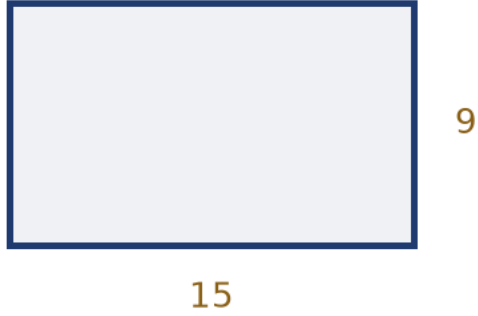
- 58 A triangle has sides 9 cm, 12 cm, and 15 cm, as shown. Is it a right triangle?



- a. Only if it is also isosceles
 - b. Yes — a right triangle
 - c. Cannot be determined
 - d. No — not a right triangle
- 59 A 15 m ladder leans against a wall with its foot 9 m from the wall's base. How high up the wall does the ladder reach?
- a. 13
 - b. 10
 - c. 12
 - d. 18

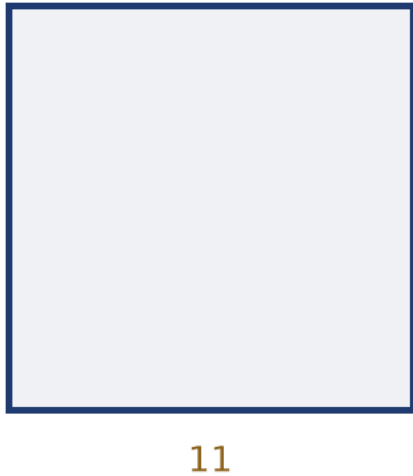
MCQ — Perimeter and area of rectangles and squares

60 Find the perimeter and area of the rectangle shown.



- a. $P = 48 \text{ cm}$, $A = 135 \text{ cm}^2$
- b. $P = 48 \text{ cm}$, $A = 24 \text{ cm}^2$
- c. $P = 135 \text{ cm}$, $A = 48 \text{ cm}^2$
- d. $P = 24 \text{ cm}$, $A = 135 \text{ cm}^2$

- 61 Find the perimeter and area of the square shown.



- a. $P = 44 \text{ cm}$, $A = 110 \text{ cm}^2$
 - b. $P = 44 \text{ cm}$, $A = 121 \text{ cm}^2$
 - c. $P = 22 \text{ cm}$, $A = 121 \text{ cm}^2$
 - d. $P = 121 \text{ cm}$, $A = 44 \text{ cm}^2$
- 62 A rectangle has perimeter 64 cm and length 20 cm. Find its width and area.
- a. $w = 12 \text{ cm}$, $A = 32 \text{ cm}^2$
 - b. $w = 12 \text{ cm}$, $A = 240 \text{ cm}^2$
 - c. $w = 44 \text{ cm}$, $A = 240 \text{ cm}^2$
 - d. $w = 22 \text{ cm}$, $A = 440 \text{ cm}^2$

63 A square has area 225 cm^2 . Find its side length and perimeter.

a. $s = 15 \text{ cm}$, $P = 30 \text{ cm}$

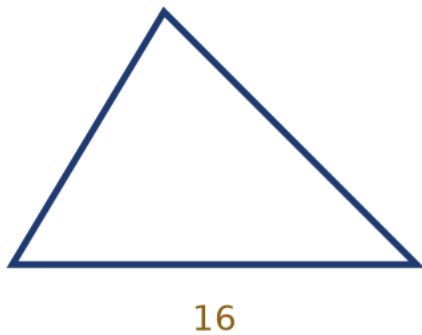
b. $s = 225 \text{ cm}$, $P = 900 \text{ cm}$

c. $s = 15 \text{ cm}$, $P = 60 \text{ cm}$

d. $s = 45 \text{ cm}$, $P = 60 \text{ cm}$

MCQ — Area of triangles and parallelograms

64 The triangle shown has base 16 cm and height 10 cm. Find its area.



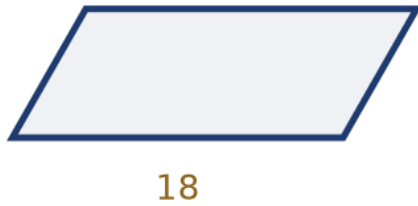
a. 160

b. 26

c. 80

d. 40

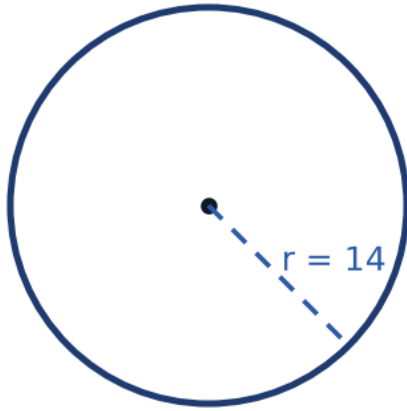
- 65 The parallelogram shown has base 18 cm and height 7 cm. Find its area.



- a. 252
 - b. 126
 - c. 63
 - d. 25
- 66 A triangle has area 84 cm^2 and height 12 cm. Find its base.
- a. 14
 - b. 1,008
 - c. 28
 - d. 7
- 67 A parallelogram has area 150 cm^2 and base 15 cm. Find its height.
- a. 20
 - b. 10
 - c. 2,250
 - d. 5

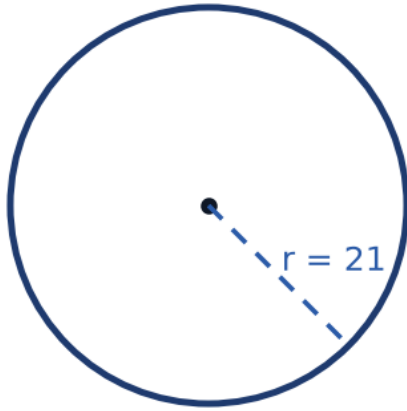
MCQ — Circles: circumference and area

- 68 Find the circumference of the circle shown, using $\pi = \frac{22}{7}$.



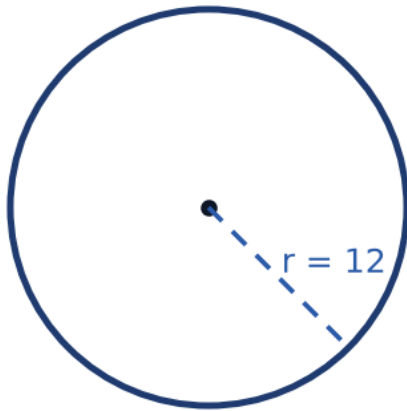
- a. 44
- b. 616
- c. 88
- d. 176

- 69 Find the area of the circle shown, using $\pi = \frac{22}{7}$.



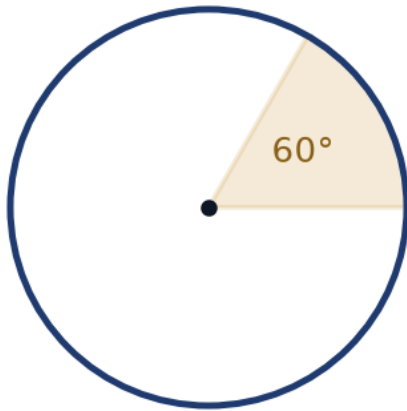
- a. 132
- b. 2,772
- c. 693
- d. 1,386

70 Find the circumference of the circle shown, in terms of π .



- a. 48π cm
- b. 24π cm
- c. 12π cm
- d. 144π cm

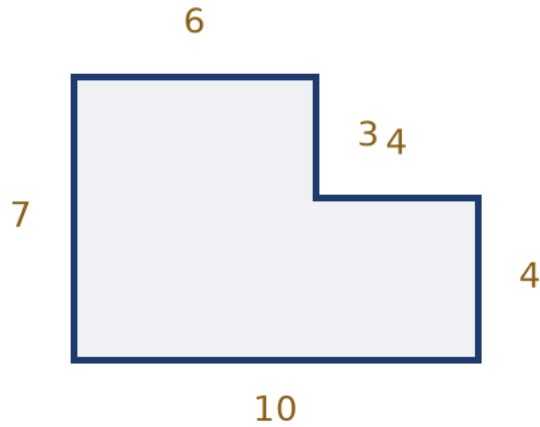
- 71 Find the area of the shaded 60° sector shown, in terms of π .



- a. $81\pi \text{ cm}^2$
 - b. $27\pi \text{ cm}^2$
 - c. $13.5\pi \text{ cm}^2$
 - d. $4.5\pi \text{ cm}^2$
- 72 A circle has area $144\pi \text{ cm}^2$. Find its radius.
- a. 6
 - b. 24
 - c. 12
 - d. 72

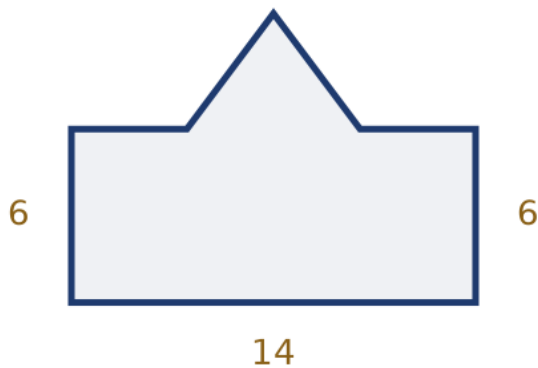
MCQ — Composite figures

- 73 Find the area of the L-shaped room shown (all measurements in meters).



- a. 70
 - b. 58
 - c. 46
 - d. 82
- 74 A garden is L-shaped: an 18 m by 8 m rectangle with a 6 m by 3 m rectangle removed from one corner. Find its area.
- a. 108
 - b. 126
 - c. 162
 - d. 138

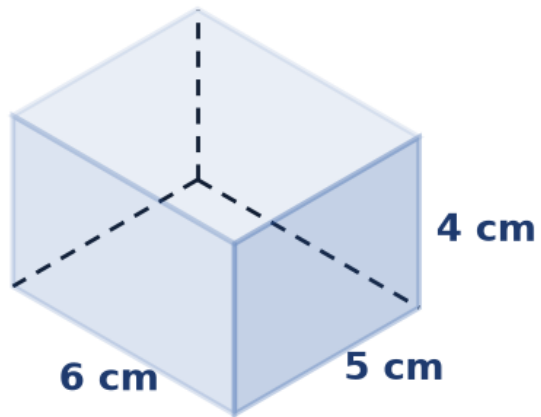
- 75 Find the area of the shape shown: a 14 cm by 6 cm rectangle with a right triangle of base 6 cm and height 4 cm attached to one side.



- a. 108
- b. 90
- c. 96
- d. 72
- 76 A rectangular lawn 20 m by 12 m has a circular flower bed of radius 3.5 m removed from it. Find the remaining lawn area, using $\pi = \frac{22}{7}$.
- a. 196
- b. 278.5
- c. 240
- d. 201.5

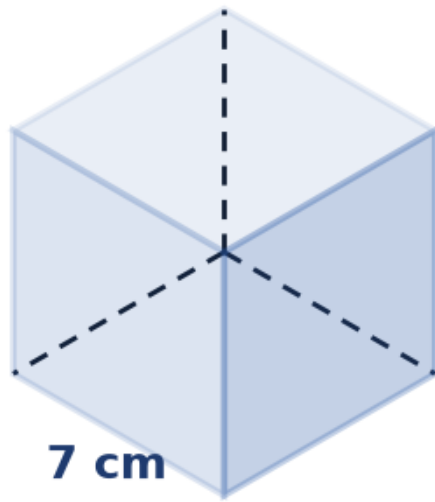
MCQ — Volume and surface area

77 Find the volume of the rectangular box shown.



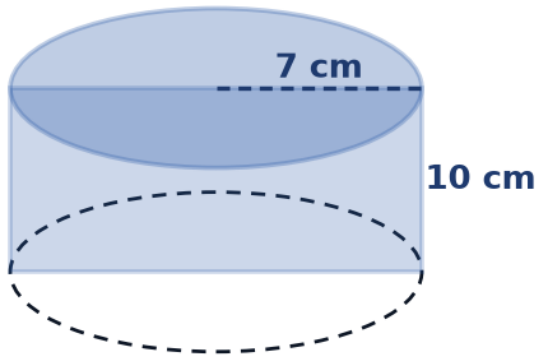
- a. 120
- b. 60
- c. 15
- d. 30

78 Find the volume and surface area of the cube shown.



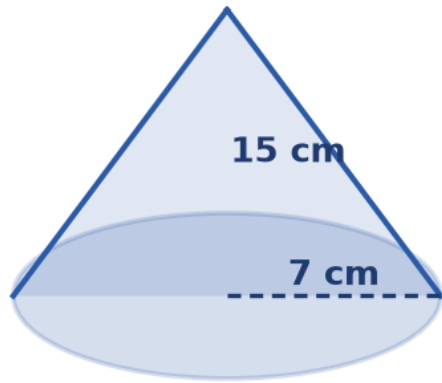
- a. $V = 343 \text{ cm}^3$, $SA = 49 \text{ cm}^2$
- b. $V = 49 \text{ cm}^3$, $SA = 294 \text{ cm}^2$
- c. $V = 294 \text{ cm}^3$, $SA = 343 \text{ cm}^2$
- d. $V = 343 \text{ cm}^3$, $SA = 294 \text{ cm}^2$

- 79 Find the volume of the cylinder shown, using $\pi = \frac{22}{7}$.



- a. 1,540
- b. 770
- c. 440
- d. 3,080

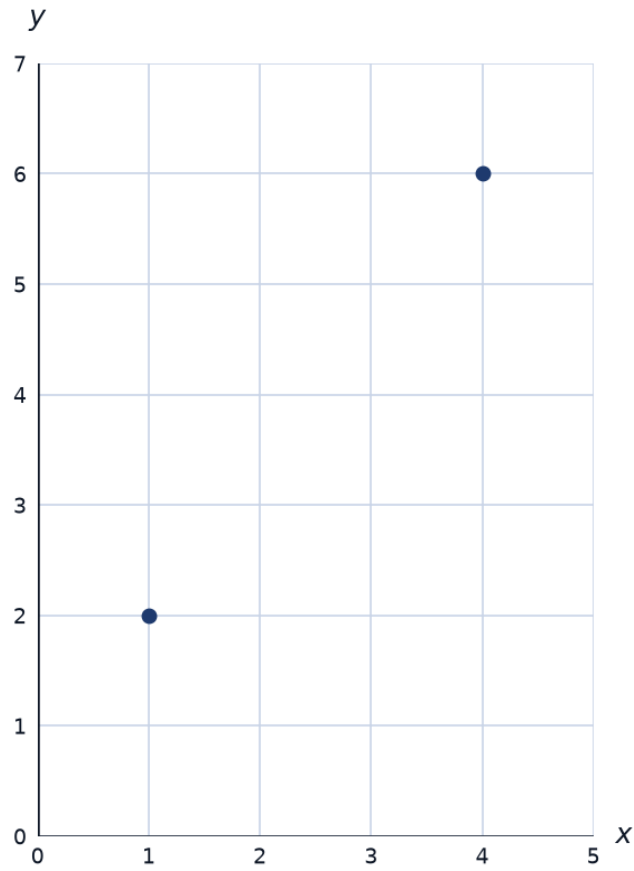
- 80 Find the volume of the cone shown, using $\pi = \frac{22}{7}$.



- a. 770
- b. 2,310
- c. 1,540
- d. 385
- 81 A rectangular prism has volume 360 cm^3 , length 12 cm, and width 5 cm. Find its height.
- a. 6
- b. 15
- c. 72
- d. 30

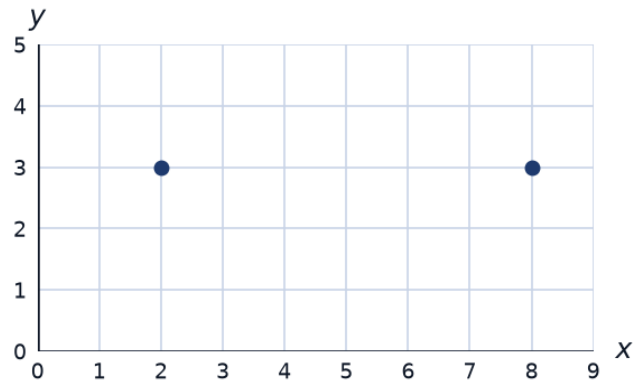
MCQ — Coordinate geometry: distance and midpoint

82 Find the distance between the two points shown.



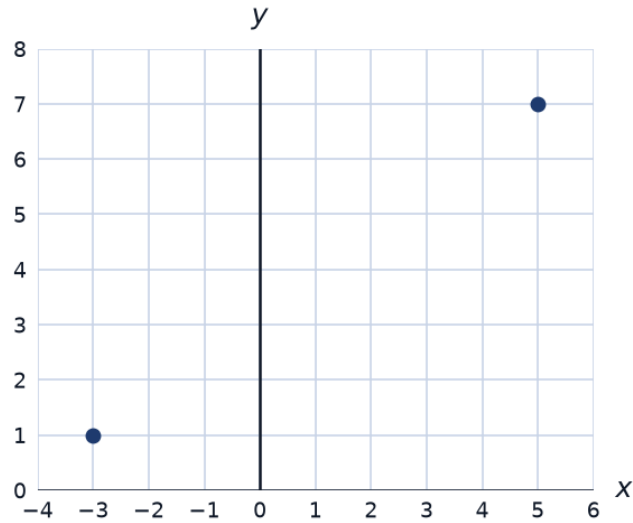
- a. 5
- b. 25
- c. 7
- d. $\sqrt{7}$

- 83 Find the midpoint of the segment joining the two points shown.



- a. (6, 3)
 - b. (3, 5)
 - c. (10, 6)
 - d. (5, 3)
- 84 Find the distance between the points (0, 0) and (6, 8).
- a. 14
 - b. 10
 - c. 7
 - d. 48

- 85 Find the midpoint of the segment joining the two points shown.



- a. $(-1, -4)$
- b. $(4, 1)$
- c. $(2, 8)$
- d. $(1, 4)$

MCQ — Similar figures and scale factor

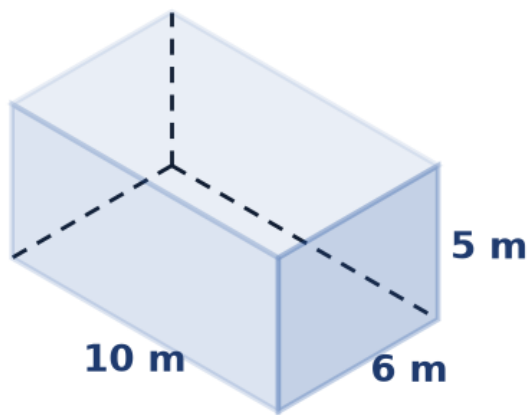
- 86 Two similar triangles have a scale factor of 3 : 5. If the smaller triangle's perimeter is 24 cm, find the larger triangle's perimeter.
- a. 14.4
 - b. 72
 - c. 64
 - d. 40

- 87 A model car is built at a scale of 1 : 24. If the model is 8 cm long, find the actual car's length in meters.
- a. 19.2
 - b. 0.192
 - c. 1.92
 - d. 24
- 88 Two similar rectangles have areas in ratio 4 : 9. Find the ratio of their side lengths.
- a. 4 : 9
 - b. 2 : 3
 - c. 8 : 18
 - d. 16 : 81
- 89 A photo is enlarged by a scale factor of 2.5. If the original photo is 6 cm wide, find the width of the enlargement.
- a. 8.5
 - b. 12.5
 - c. 3.6
 - d. 15

MCQ — Word problems

- 90 A rectangular pool measuring 25 m by 12 m is surrounded by a deck 2 m wide. Find the area of the deck alone.
- a. 74
 - b. 300
 - c. 164
 - d. 464

- 91 Tiling costs 38 riyals per square meter. Find the total cost to tile a rectangular room measuring 9 m by 6 m.
- a. 1,026 SAR
 - b. 4,104 SAR
 - c. 342 SAR
 - d. 2,052 SAR
- 92 A water tank shaped like the rectangular box shown has these dimensions in meters. Find its capacity in cubic meters.



- a. 150
- b. 60
- c. 300
- d. 21

Multiple choice — easy

- 93 A right triangle has legs 11 cm and 60 cm. Find the hypotenuse.
- a. 61
 - b. 59
 - c. 71
 - d. 63
- 94 A right triangle has legs 44 cm and 240 cm. Find the hypotenuse.
- a. 244
 - b. 242
 - c. 246
 - d. 284
- 95 A right triangle has legs 5 cm and 12 cm. Find the hypotenuse.
- a. 15
 - b. 11
 - c. 13
 - d. 17
- 96 A right triangle has legs 33 cm and 180 cm. Find the hypotenuse.
- a. 183
 - b. 181
 - c. 213
 - d. 185

97 A right triangle has legs 10 cm and 24 cm. Find the hypotenuse.

a. 34

b. 24

c. 28

d. 26

98 A right triangle has legs 12 cm and 16 cm. Find the hypotenuse.

a. 18

b. 20

c. 28

d. 22

99 A right triangle has legs 36 cm and 160 cm. Find the hypotenuse.

a. 196

b. 164

c. 162

d. 166

100 A right triangle has legs 20 cm and 48 cm. Find the hypotenuse.

a. 54

b. 52

c. 50

d. 68

101 A right triangle has legs 3 cm and 4 cm. Find the hypotenuse.

- a. 3
- b. 9
- c. 7
- d. 5

102 A right triangle has legs 9 cm and 40 cm. Find the hypotenuse.

- a. 39
- b. 41
- c. 43
- d. 49

103 A right triangle has legs 21 cm and 72 cm. Find the hypotenuse.

- a. 77
- b. 93
- c. 75
- d. 73

104 A right triangle has legs 12 cm and 35 cm. Find the hypotenuse.

- a. 37
- b. 47
- c. 39
- d. 35

105 A right triangle has legs 27 cm and 120 cm. Find the hypotenuse.

- a. 121
- b. 125
- c. 123
- d. 147

106 A right triangle has legs 24 cm and 45 cm. Find the hypotenuse.

- a. 49
- b. 53
- c. 51
- d. 69

107 A right triangle has legs 48 cm and 140 cm. Find the hypotenuse.

- a. 146
- b. 150
- c. 188
- d. 148

Multiple choice — medium

108 A rectangle has length 22 cm and width 21 cm. Find its perimeter and area.

- a. 86, 468
- b. 462, 86
- c. 90, 462
- d. 86, 462

109 A rectangle has length 9 cm and width 8 cm. Find its perimeter and area.

- a. 34, 78
- b. 38, 72
- c. 34, 72
- d. 72, 34

110 A rectangle has length 17 cm and width 6 cm. Find its perimeter and area.

- a. 46, 102
- b. 50, 102
- c. 46, 108
- d. 102, 46

111 A rectangle has length 23 cm and width 5 cm. Find its perimeter and area.

- a. 56, 121
- b. 56, 115
- c. 115, 56
- d. 60, 115

112 A rectangle has length 7 cm and width 5 cm. Find its perimeter and area.

- a. 24, 41
- b. 35, 24
- c. 28, 35
- d. 24, 35

113 A rectangle has length 6 cm and width 4 cm. Find its perimeter and area.

- a. 24, 24
- b. 20, 24
- c. 20, 30
- d. 24, 20

114 A rectangle has length 10 cm and width 4 cm. Find its perimeter and area.

- a. 28, 46
- b. 40, 28
- c. 28, 40
- d. 32, 40

115 A rectangle has length 15 cm and width 4 cm. Find its perimeter and area.

- a. 42, 60
- b. 38, 66
- c. 60, 38
- d. 38, 60

116 A rectangle has length 21 cm and width 3 cm. Find its perimeter and area.

- a. 52, 63
- b. 48, 69
- c. 48, 63
- d. 63, 48

117 A rectangle has length 4 cm and width 3 cm. Find its perimeter and area.

- a. 14, 12
- b. 14, 18
- c. 12, 14
- d. 18, 12

118 A rectangle has length 12 cm and width 6 cm. Find its perimeter and area.

- a. 36, 72
- b. 36, 78
- c. 72, 36
- d. 40, 72

119 A rectangle has length 25 cm and width 13 cm. Find its perimeter and area.

- a. 325, 76
- b. 76, 331
- c. 80, 325
- d. 76, 325

120 A rectangle has length 11 cm and width 5 cm. Find its perimeter and area.

- a. 32, 61
- b. 32, 55
- c. 55, 32
- d. 36, 55

121 A rectangle has length 24 cm and width 17 cm. Find its perimeter and area.

- a. 86, 408
- b. 82, 414
- c. 408, 82
- d. 82, 408

122 A rectangle has length 21 cm and width 14 cm. Find its perimeter and area.

- a. 294, 70
- b. 70, 294
- c. 70, 300
- d. 74, 294

Multiple choice — hard

123 An isosceles triangle has two equal angles of 74° each. Find the third angle.

- a. 38
- b. 74
- c. 32
- d. 26

124 An isosceles triangle has two equal angles of 59° each. Find the third angle.

- a. 59
- b. 62
- c. 68
- d. 56

- 125** A triangle has two angles of 46° and 28° . Find the third angle.
- a. 74
 - b. 106
 - c. 111
 - d. 101
- 126** An isosceles triangle has two equal angles of 64° each. Find the third angle.
- a. 52
 - b. 46
 - c. 64
 - d. 58
- 127** A triangle has two angles of 92° and 56° . Find the third angle.
- a. 27
 - b. 37
 - c. 148
 - d. 32
- 128** Find the sum of the interior angles of a pentagon (5 sides).
- a. 360
 - b. 540
 - c. 720
 - d. 900

129 Find the sum of the interior angles of a decagon (10 sides).

- a. 1620
- b. 1440
- c. 1260
- d. 1800

130 Find the sum of the interior angles of a nonagon (9 sides).

- a. 1440
- b. 1080
- c. 1260
- d. 1620

131 Find the sum of the interior angles of a heptagon (7 sides).

- a. 1260
- b. 900
- c. 720
- d. 1080

132 Find the sum of the interior angles of an octagon (8 sides).

- a. 1080
- b. 1440
- c. 1260
- d. 900

133 A circle has radius 49 cm. Find its area, using $\pi = \frac{22}{7}$.

a. 7532

b. 308

c. 7546

d. 7560

134 A circle has radius 28 cm. Find its area, using $\pi = \frac{22}{7}$.

a. 2450

b. 2478

c. 176

d. 2464

135 A circle has radius 14 cm. Find its circumference, using $\pi = \frac{22}{7}$.

a. 84

b. 616

c. 88

d. 92

136 A circle has radius 35 cm. Find its area, using $\pi = \frac{22}{7}$.

a. 3864

b. 220

c. 3850

d. 3836

137 A circle has radius 14 cm. Find its area, using $\pi = \frac{22}{7}$.

a. 616

b. 630

c. 602

d. 88